

Wherever you see **red characters**, these need to be replaced by your information. This includes the **< >** symbols!

Creating a new variable that combines two variables

```
<NAME OF DATASET> <- <NAME OF DATASET> |>
  mutate(<NEW VAR NAME> = paste0(<NAME of VARIABLE 1>,
                                <NAME of VARIABLE 2>,
                                ...))
```

We used this to create a treatment variable from multiple factor variables:

```
snackdat <- snackdat |>
  mutate(treatment = paste0("Ad: ", adexposure,
                            ", Snack: ", snack))
```

In this example above, I added extra strings "Ad: " and "Snack: " for clarity, but it wasn't needed.

ANOVA F-Test Adjusting for Confounding Additional Variables

```
aov3(<NAME OF RV> ~ <NAME OF EV> + <NAME OF ADDITIONAL VARIABLE> + ...,
     Data = <NAME OF DATASET>, BETA = FALSE, Resid = FALSE)
```

Runs a correctly adjusted theory-based ANOVA F-test. Output is the adjusted ANOVA table.

Note: The only things you need to change for an analysis is the **formula** and the name of the dataset.

ANOVA F-Test Adjusting for Independent Additional Variables

```
mymod <- lm(<NAME OF RV> ~ <NAME OF EV> + <NAME OF ADDITIONAL VARIABLE> + ...,
           data = <NAME OF DATASET>)
anova(mymod)
```

Note that this is THE SAME as with just one explanatory variable. The only difference is using the appropriate **formula** input that now includes additional variables. **Caution:** This only gives the correct output if the additional variables are all independent of each other and the explanatory variable. If you are not sure, use the method above and it is always right.

Runs a theory-based ANOVA F-test. Output is the ANOVA table.

Note that you can call **mymod** anything you want but then you must use the same name of that model that you created in the `anova()` function.

See Linear Regression Model Output

```
mymod <- lm(<NAME OF RV> ~ <NAME OF EV> + <NAME OF ADDITIONAL VARIABLE> + ...,
           data = <NAME OF DATASET>)
summary(mymod)
```

Output will be the estimated coefficients for the model and other information like R² etc.